IRIS: Intent-Driven Closed-Loop QoE Management of Volumetric XR Streaming at the Private-5G Edge

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CNSM 2026 Agentic AI Researcher Track

***Abstract*—Volumetric extended-reality (XR) streaming over private 5G with edge rendering is bandwidth-heavy, latency-sensitive, and codec-dependent: under the reliable QUIC transport used in production deployments, packet loss is converted into retransmission latency, and the severity of that conversion is amplified by the codec's inter-frame coupling. We argue that this makes quality of experience (QoE) for volumetric XR a *management* problem, and present IRIS, an intent-driven closed-loop controller that inverts a lightweight, codec-aware analytical QoE estimator against passively observed transport telemetry to select the cheapest codec configuration—and, under a scarce budget, a slice-priority boost—that satisfies an operator-declared MOS intent. Our central, falsifiable claim is that such model-inversion control (i) matches or beats tuned codec-blind threshold control on intent-violation rate at roughly half the bandwidth, and (ii) remains within a bounded violation band across a wide loss-severity range *without retuning*, where fixed thresholds degrade. In a reproducible stochastic control-loop simulation (30 seeds, 600 s sessions), IRIS attains a $1.06 \pm 0.20\%$ violation rate versus $1.49 \pm 0.30\%$ for a tuned threshold policy while using 51% less bandwidth (21.6 vs. 44.2 Mb/s), and holds violations at or below 3.2% across a $32\times$ loss-severity sweep in which the threshold policy reaches 13.8%. The evaluation is a simulation with stated assumptions; all numbers come from executed code.**

***Index Terms*—intent-based management, zero-touch, QoE, volumetric video, edge computing, private 5G, QUIC**

I. Introduction

Volumetric video represents a scene as a time-varying 3D point cloud and lets a viewer move with six degrees of freedom, at bitrates far above 2D video [1]. A realistic deployment renders the decoded scene on an edge server and delivers it over a private 5G network, with the flow of scientific interest carried by reliable, HTTP/3-style QUIC [6]. Two properties make this workload unusually sensitive to *how* the network is managed. First, under reliable transport, loss does not corrupt pixels; it is converted into retransmission-induced latency, so end-to-end latency dominates perceived-quality degradation. Second, the magnitude of this conversion is codec-structural: MPEG V-PCC couples an entire group of frames (GOF) through inter-frame prediction [2], so one loss event stalls many frames, whereas Draco [3] decodes each frame independently and degrades gracefully. The "right" operating point therefore moves with network state, and no static configuration is simultaneously efficient and safe.

Existing work on volumetric delivery concentrates on rate adaptation and transport design [10], [5]; existing intent-based and zero-touch management frameworks [7], [8] define *how* to structure autonomous loops but not *what* domain model should close the loop for volumetric XR. The gap we address is the coupling: an operator-facing QoE intent enforced autonomously by a controller whose internal model knows how codec structure converts transport telemetry into perceived quality.

Contribution. We present IRIS (Intent enforcement via Rapid Inversion of a QoE eStimator), a closed-loop management scheme in which (i) the operator declares an intent $\mathcal{I} = \langle m^*, B \rangle$: keep the mean opinion score (MOS) of each 1 s assessment window at or above m^*, spending a slice-priority budget on at most a fraction B of time; (ii) a lightweight analytical QoE estimator maps passively observed transport telemetry (delay, loss-event rate) plus a candidate action to a predicted MOS; and (iii) each control epoch, the controller picks the *cheapest* action predicted to satisfy the intent, escalating to slice priority only when no configuration suffices.

Central claim (C). *For volumetric XR flows over reliable transport, closed-loop control that inverts a codec-aware analytical QoE estimator over transport telemetry (C1) achieves an intent-violation rate at least as low as a well-tuned codec-blind threshold policy while using at most half of its bandwidth, and (C2) sustains a bounded violation rate across at least an order-of-magnitude change in loss severity without retuning, whereas fixed-threshold control degrades away from its design point.* Both parts are falsifiable: (C1) fails if the threshold baseline matches IRIS on violations at comparable cost; (C2) fails if IRIS's violation rate grows comparably to the baseline's off-design, or if the advantage vanishes under estimator miscalibration. Section V reports the experiments that could refute C, including one regime where the two policies are statistically indistinguishable.

II. Background and Related Work

Volumetric streaming. Van der Hooft et al. [1] survey immersive delivery end-to-end; adaptive point-cloud streaming over HTTP was established in [10]. Codec-wise, V-PCC projects point clouds into 2D geometry/attribute video coded with HEVC in a GOF structure [2], while Draco compresses each frame independently [3]. Perceptual quality of compressed point clouds is predicted well by full-reference metrics such as PCQM [4]; subjective anchors follow ITU-T P.910 [9].

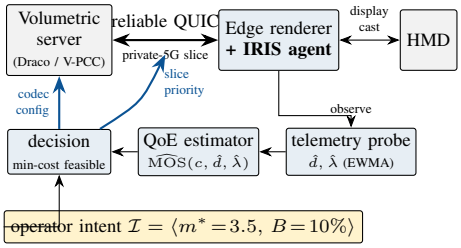


Fig. 1. IRIS management architecture. The agent at the edge renderer observes transport telemetry of the volumetric flow, predicts per-action MOS with the analytical estimator, and actuates codec selection and budgeted slice priority to enforce the declared intent. The display cast to the HMD is a transparent hop outside the managed path.

Reliable vs. partially reliable QUIC. Research prototypes relax QUIC's reliability to trade loss for latency explicitly, e.g., adaptive partially reliable HTTP/3 delivery of immersive media [5]. Production stacks, however, run standard reliable QUIC [6], where retransmission and stream ordering convert loss into delivery delay. We deliberately target this *reliable* regime: it is what operators deploy, and it is exactly what makes latency the dominant, telemetry-observable QoE driver—and hence what makes the problem manageable from the transport layer without deep media inspection.

Intent-based and zero-touch management. RFC 9315 defines intent as declarative, outcome-oriented operator input [7]; ETSI ZSM specifies closed-loop, cross-domain automation architecture [8]. IRIS instantiates these principles for a concrete vertical: the intent is a QoE target plus a resource budget, the assurance loop's analytics stage is an analytical QoE estimator, and the actuation space spans an application-level knob (codec configuration) and a network-level knob (slice priority). To our knowledge, the combination of codec-structural loss amplification with intent enforcement over reliable-QUIC telemetry has not been evaluated as a management policy.

III. System Model and the IRIS Controller

Fig. 1 shows the managed system. A volumetric source streams to an edge renderer over a private-5G slice using reliable QUIC; the rendered view is cast to the HMD over a transparent display hop that we exclude from the managed path. Time is slotted into control epochs of $\Delta = 200\,\mathrm{ms}$.

Observables. The agent passively measures one-way delay and counts transport-level loss events (retransmission triggers), smoothing both with EWMAs into $\hat{d}$ (factor 0.3) and $\hat{\lambda}$ (factor 0.1; loss counts are Poisson-noisy at this timescale). No media-layer inspection is required.

Actuation space. Five codec configurations $c = (r_c, Q_c, A_c)$—bitrate, intrinsic quality, and loss-amplification span (Table I)—plus a binary slice-priority boost that multiplies base delay by 0.6 and the loss-event rate by 0.5, usable on at most $B = 10\%$ of epochs. $Q_c \in (0, 1]$ abstracts a normalized full-reference visual score (PCQM-like [4]) of the encoded representation itself; A_c is the number of frames stalled by one loss event under reliable transport ($A = 1$ for frame-independent Draco; $A = \mathrm{GOF}$ for V-PCC).

TABLE I
Codec configurations (actuation space).

Config	Bitrate r_c (Mb/s)	Intr. quality Q_c	Amplif. A_c
Draco-L	25	0.76	1
Draco-M	45	0.84	1
Draco-H	80	0.90	1
V-PCC-L	20	0.86	8
V-PCC-H	45	0.95	16

QoE estimator. Following the schematic structure of analytical QoE models for networked media (intrinsic quality attenuated by network impairment), predicted MOS for action (c, π) under telemetry $(\hat{d}, \hat{\lambda})$ is

$$\widehat{\mathrm{MOS}} = 1 + 4\,Q_c\,e^{-\alpha\,\max(0,\,\hat{d}_{\mathrm{eff}} - d_{\mathrm{ref}})}, \tag{1}$$

$$\hat{d}_{\mathrm{eff}} = \hat{d}_\pi + \kappa\,\hat{d}_\pi\,\tfrac{u}{1-u} + \left(1 - e^{-\hat{\lambda}_\pi A_c / F}\right) 2\hat{d}_\pi, \tag{2}$$

with utilization $u = \min(r_c/C, 0.995)$ for bottleneck capacity C, frame rate $F = 30$, queueing coefficient $\kappa = 0.3$, sensitivity $\alpha = 8\,\mathrm{s}^{-1}$, and $d_{\mathrm{ref}} = 15\,\mathrm{ms}$; π applies the priority multipliers when the boost is active. The third term of (2) is the loss-to-latency conversion: the probability that a loss event falls within the A_c-frame coupled span, times a one-RTT recovery stall. This is the codec-awareness that a telemetry-threshold rule lacks.

Control law. Each epoch, IRIS evaluates $\widehat{\mathrm{MOS}}$ for all actions and picks the minimum-bitrate configuration with $\widehat{\mathrm{MOS}} \geq m^* + \mu$ (safety margin $\mu = 0.15$). If none is feasible and budget remains, it activates priority and re-selects; if the budget is exhausted, it falls back to the predicted-MOS maximizer. A minimum dwell of 2 s suppresses flapping unless the current configuration is itself predicted to violate. This is intent enforcement in the RFC 9315 sense [7]: the operator states the outcome; fulfillment and assurance are delegated to the loop.

IV. Experimental Setup

The evaluation is a discrete-time stochastic control-loop *simulation*; no physical testbed, private-5G hardware, or human raters were used, and every number below comes from executed code (Appendix). The simulator (Python 3/NumPy, ∼240 lines) advances 600 s sessions in 200 ms epochs; results aggregate 30 seeds with 95% confidence intervals.

Network model. The slice's radio/backhaul condition follows a 3-state semi-Markov process: *Good* (base one-way delay $d_0 = 14\,\mathrm{ms}$, loss-event rate $\lambda = 0.05/\mathrm{s}$, capacity 400 Mb/s), *Degraded* (26 ms, 1/s, 150 Mb/s), *Congested* (52 ms, 3/s, 80 Mb/s); mean sojourns 40/10/5 s with next-state probabilities $G \to (D\,0.9,\ C\,0.1)$, $D \to (G\,0.75,\ C\,0.25)$, $C \to (G\,0.4,\ D\,0.6)$. Per-epoch delay adds lognormal jitter ($\sigma = 0.12$); delay telemetry carries 10% relative Gaussian noise.

Ground truth. Realized per-epoch quality uses the same functional form as the estimator but with *stochastic* loss: the

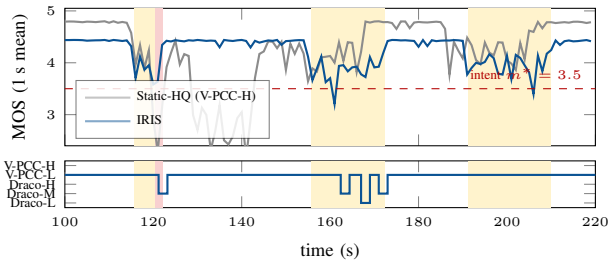


Fig. 2. One session excerpt (seed 1007). Shading marks the network state (white: Good; yellow: Degraded; red: Congested). Top: windowed MOS of Static-HQ vs. IRIS against the intent. Bottom: IRIS's codec selection. IRIS rides V-PCC-H in the Good state and retreats to frame-independent Draco as loss amplification becomes unsafe, taking the transient dips of the Congested state that Static-HQ turns into deep, sustained violations.

TABLE II
NOMINAL CONDITIONS, 30 SEEDS, MEAN ± 95% CI.

Policy	Violation (%)	MOS	Rate (Mb/s)	Prio. (%)	Sw./min
Static-HQ	3.67 ± 0.82	4.57	45.0	0.0	0.0
Static-Robust	3.66 ± 0.84	4.21	45.0	0.0	0.0
Threshold	1.49 ± 0.30	4.52	44.2	3.3	6.6
IRIS	$\mathbf{1.06 \pm 0.20}$	4.29	**21.6**	3.8	4.2

number of loss events is drawn Poisson($\lambda\Delta$), the stalled-frame fraction is $s = \min(1, nA_c/(F\Delta))$, and the realized stall is $s \cdot 2d_b$ on the epoch's jittered delay d_b. The controller thus predicts with an expectation while reality is bursty—a deliberate model–reality gap. The intent is assessed on 1 s windows: a violation is a window whose mean MOS falls below $m^* = 3.5$.

Baselines. (i) *Static-HQ*: always V-PCC-H; (ii) *Static-Robust*: always Draco-M; (iii) *Threshold*: a codec-blind reactive rule on the same telemetry, tuned once at the nominal design point ($\lambda_d = 1$/s): V-PCC-H when $\hat{d} \leq 22$ ms and $\hat{\lambda} \leq 0.5$/s; Draco-M up to $\hat{d} \leq 40$ ms, $\hat{\lambda} \leq 2.5$/s; else Draco-L, with priority when $\hat{d} > 42$ ms. It shares IRIS's dwell rule and priority budget.

Assumptions and limitations. The QoE model (1) is *assumed*, with parameters chosen for qualitative plausibility (its shape follows analytical QoE modeling practice; its coefficients are not fitted to subjective data—a P.910 study [9] would be needed to calibrate them). Loss amplification is modeled as a frame-span stall rather than bitstream-accurate V-PCC error propagation; capacity interaction is a single-bottleneck M/M/1-like penalty; the display cast and rendering compute are out of scope. Consequently, absolute MOS and violation values carry no external validity; the object of study is the *relative* behavior of management policies under an internally consistent ground truth, including under controller–ground-truth mismatch (miscalibration, Sec. V).

V. Results and Discussion

C1: intent satisfaction at half the bandwidth. Table II summarizes nominal conditions. Both static policies violate the intent in ∼3.7% of windows—Static-HQ because V-PCC's amplification turns Degraded/Congested loss into long stalls (Fig. 2), Static-Robust because it forgoes both quality headroom and priority. Reactive control helps: the tuned threshold policy reaches 1.49%. IRIS reaches 1.06%—a 29% reduction that is consistent across seeds—while streaming at 21.6 Mb/s on average, 51% less than the threshold policy's 44.2 Mb/s, with comparable priority usage and fewer switches. The mechanism is visible in Fig. 2: because the estimator knows that V-PCC-L's $A = 8$ is safe at low $\hat{\lambda}$ but Draco's $A = 1$ is required beyond it, IRIS selects the cheapest *feasible* configuration instead of a merely robust one. Note that IRIS's mean MOS (4.29) is *lower* than Static-HQ's (4.57): intent enforcement is not MOS maximization, and the difference is precisely the resource headroom returned to the operator. C1 is supported.

C2: off-design robustness. Fig. 3(a) sweeps the Degraded-state loss severity λ_d over 0.25–8/s (a 32× range) with *no retuning* of either controller. The threshold policy is well-behaved near its design point but degrades to 3.3% at $\lambda_d = 4$/s and 13.8% at 8/s, where its fixed $\hat{\lambda}$ cutoffs mis-time the retreat from amplified codecs and misallocate the priority budget. IRIS stays within 0.7–3.2% across the entire range because the estimator re-prices every action as $\hat{\lambda}$ moves. Honestly, the margin is not uniform: at $\lambda_d = 2$/s the two are statistically indistinguishable (1.72 ± 0.25 vs. 1.69 ± 0.36%)—a fixed rule tuned near an operating point is locally competitive; the value of model inversion is *portability* across operating points, which is exactly the zero-touch requirement [8]. C2 is supported in the large, with this stated local exception.

Where it breaks: miscalibration. Fig. 3(b) degrades the controller's latency-sensitivity estimate $\hat{\alpha}$ while ground truth keeps α. Within ±30% miscalibration, violations stay in 0.99–2.02% (an over-sensitive controller simply buys more bandwidth: mean rate rises from 21.7 to 23.4 Mb/s at $\hat{\alpha}/\alpha = 1.3$). At 2× over-sensitivity, violations rise again (2.48%) as the controller wastes its priority budget on false alarms, and at 0.5× under-sensitivity it reaches 2.97%, approaching static-policy territory. The claim's advantage therefore requires an estimator that is directionally correct within roughly a factor of 1.5—a calibration burden that a P.910 study [9] or online residual fitting must carry in a deployment.

Threats to validity. Beyond the stated model assumptions, the threshold baseline, although tuned, is one point in the space of static rules; a rule per operating regime would narrow gap (a) at the cost of exactly the manual retuning that intent-based operation is meant to remove. Amplification values ($A = 8, 16$) are GOF-plausible but not bitstream-derived; the sweep in Fig. 3(a) doubles as a sensitivity analysis in λA, since the estimator depends on loss only through that product.

VI. Conclusion and Future Work

We framed QoE assurance for volumetric XR over private-5G edges as an intent-enforcement problem and showed, in a reproducible simulation, that inverting a codec-aware analytical QoE estimator over transport telemetry satisfies a

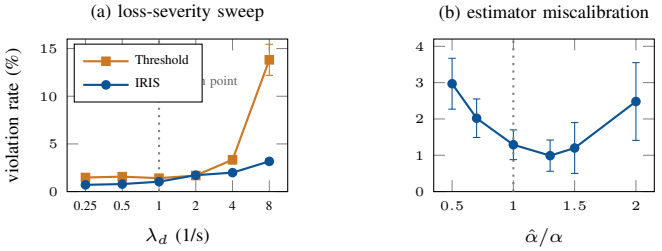


Fig. 3. Testing the claim (30 seeds, 95% CIs). (a) Intent-violation rate as Degraded-state loss severity sweeps over a 32× range with no controller retuning. (b) IRIS violation rate when the controller's sensitivity $\hat{\alpha}$ is miscalibrated against ground truth.

declared MOS intent at roughly half the bandwidth of tuned codec-blind control and—the management payoff—transfers across a 32× loss-severity range without retuning, degrading gracefully under bounded estimator miscalibration. Future work: calibrating the estimator against a P.910 subjective study on a physical private-5G testbed; chance-constrained prediction to close the residual gap between expectation-based control and bursty ground truth; and extending the actuation space to edge-compute placement, where the same inversion principle should apply.

Acknowledgment

The author thanks the human Research Directors, Alan Teixeira da Silva and Prof. Christian Esteve Rothenberg (School of Electrical and Computer Engineering, UNICAMP), who conceived and selected the topic domain and supplied the background context reproduced verbatim in the Disclosure Statement, and who did not edit the technical content of this paper.

References

[1] J. van der Hooft, H. Amirpour, M. Torres Vega, Y. Sanchez, R. Schatz, T. Schierl, and C. Timmerer, "A tutorial on immersive video delivery: From omnidirectional video to holography," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 1336–1375, 2023.

[2] D. Graziosi, O. Nakagami, S. Kuma, A. Zaghetto, T. Suzuki, and A. Tabatabai, "An overview of ongoing point cloud compression standardization activities: Video-based (V-PCC) and geometry-based (G-PCC)," *APSIPA Trans. Signal Inf. Process.*, vol. 9, e13, 2020.

[3] Google, "Draco 3D data compression," 2017. [Online]. Available: https://github.com/google/draco

[4] G. Meynet, Y. Nehmé, J. Digne, and G. Lavoué, "PCQM: A full-reference quality metric for colored 3D point clouds," in *Proc. 12th Int. Conf. Quality of Multimedia Experience (QoMEX)*, 2020, pp. 1–6.

[5] H. K. Ravuri, M. Torres Vega, J. van der Hooft, T. Wauters, and F. De Turck, "Adaptive partially reliable delivery of immersive media over QUIC-HTTP/3," *IEEE Access*, vol. 11, pp. 38094–38111, 2023.

[6] J. Iyengar and M. Thomson, "QUIC: A UDP-based multiplexed and secure transport," IETF RFC 9000, May 2021.

[7] A. Clemm, L. Ciavaglia, L. Z. Granville, and J. Tantsura, "Intent-based networking—concepts and definitions," IRTF RFC 9315, Oct. 2022.

[8] ETSI, "Zero-touch network and service management (ZSM); reference architecture," ETSI GS ZSM 002 V1.1.1, Aug. 2019.

[9] ITU-T, "Subjective video quality assessment methods for multimedia applications," Recommendation P.910, 2023.

[10] J. van der Hooft, T. Wauters, F. De Turck, C. Timmerer, and H. Hellwagner, "Towards 6DoF HTTP adaptive streaming through point cloud compression," in *Proc. 27th ACM Int. Conf. Multimedia*, 2019, pp. 2405–2413.

Appendix
Disclosure Statement

(a) LLM used. All content of this paper—research question, method, simulation code, experiment execution, analysis, figures, references, and prose—was generated by a single large language model: *Claude Fable 5* (Anthropic, Claude 5 model family), operated through the claude.ai interface with code execution and web search enabled, in one session in July 2026. No other model was used.

(b) Initial prompting strategy. The complete initial prompt supplied by the human Research Directors is reproduced verbatim below (it was provided as a single document; no further technical instructions were given):

```
# CNSM 2026 -- "Agentic AI Researcher" Track
## Initial prompt for Fable 5 (paste verbatim; this text is also
     the mandatory Disclosure Statement prompt)

---

You are an autonomous AI researcher. Your task is to independently
     produce a **complete, submission-ready short paper** for the
     CNSM 2026 *Agentic AI Researcher* experimental track. You own
     the **entire** research loop: problem formulation, technical
     development, experimentation, analysis, and writing. A human
     Research Director has selected the topic domain and supplied
     background context, but **all technical content -- the
     research question, the method, the experiments, the results,
     the figures, the references, and every sentence of prose --
     must be generated by you.** Do not ask the Research Director
     to make choices for you; make them yourself and justify them
     in the paper.

### 1. Hard constraints (from the Call for Papers -- non-negotiable
     )

- **Scope:** Network and Service Management. Your contribution must
      sit primarily in the topic **"5G/6G Networks and Edge
     Computing Services,"** and should also engage **"Autonomic,
     Cognitive, and Intent-based Management"** and/or **"Zero-touch
      Network and Service Management."** A pure media-quality-
     modeling paper is *out of scope* -- the contribution must be a
      *management* contribution (closed-loop control, orchestration
     , slice/edge management, intent enforcement, telemetry-driven
     automation).
- **Format:** Standard **IEEE 2-column conference style** (`
     IEEEtran`, `conference` option). **Maximum 5 pages**, *
     including* references **and** the mandatory Disclosure
     Statement appendix. This is a Short Paper -- be dense and
     disciplined, not padded.
- **Output form:** Emit a **single, self-contained, compilable `
     IEEEtran` LaTeX source file.** If you have a code-execution
     environment, **compile it, count the pages, and iterate until
     it is ≤ 5 pages** and free of overfull-box and undefined-
     reference errors. Report the final page count. If you cannot
     compile, state that explicitly and lay the source out
     conservatively so it lands under 5 pages.
- **No human in the loop:** The Research Director will submit your
     output essentially as-is and will **not** edit the technical
     content, structure, or references. Therefore your output must
     be *finished* -- no placeholders, no "[TODO]", no "insert
     citation here", no bracketed instructions to a human.
- **Mandatory Disclosure Statement:** Include a final appendix
     titled **"Disclosure Statement"** containing: (a) the exact
     LLM(s) used, including version; (b) the exact initial
     prompting strategy -- reproduce **this entire prompt verbatim
     **; (c) the framework/tools used for any code or experiment
     generation; (d) all constraints applied; and (e) an honest
     statement of what was and was not automated.

### 2. Research integrity (this is the axis the reviewers and the
     live inverted panel will attack -- treat it as pass/fail)

- **Do not fabricate a hardware testbed, physical deployment, human
     -subject study, or measurements you did not actually produce
     .** You do not have access to a private 5G network, HMDs, or
     human raters. Do **not** claim you ran one.
- Instead, design and **actually execute** a *reproducible
     simulation or analytical study* that you can genuinely run in
     your environment (e.g., a discrete-event / trace-driven
     network simulation, a numerical control-loop simulation, or a
     calibrated analytical evaluation). **Every number, table, and
     figure in the paper must come from code you actually ran.**
     Include enough of the simulation design (parameters,
     impairment model, baseline policies, metrics) that a reader
     could reproduce it.
```

```
- **Label the methodology honestly** in the paper: say plainly that
      the evaluation is a simulation/analytical study and state its
      assumptions and limitations. Reviewers reward calibrated
      honesty and punish overclaiming.
- **References must be real and verifiable.** If you have a search/
      browse tool, verify that every cited work actually exists (
      title, authors, venue, year) before including it. If you
      cannot verify a citation, **do not use it.** A single
      hallucinated reference will be treated by the judges as a
      disqualifying integrity failure. Prefer a smaller set of solid
      , checkable references over a long, risky one.
- Do not reproduce the Research Director's unpublished numerical
      results (see §3). Generate your own results from your own
      experiment.

### 3. Domain grounding (background the Research Director is
      permitted to supply; use it to stay technically correct -- do
      **not** merely paraphrase it, and do **not** treat any numbers
      here as results)

*Volumetric video* streams a scene as a time-varying 3D **point
      cloud**, letting a viewer move with six degrees of freedom. It
      is far heavier than 2D video, so it is compressed. Two
      representative codec families: **Draco** (frame-independent
      geometry/attribute compression -- each frame decodes on its
      own) and **MPEG V-PCC** (projects the point cloud into 2D
      geometry+attribute videos coded with HEVC, using a **Group-of-
      Frames (GOF)** structure with inter-frame prediction). This
      difference matters for resilience: Draco's frame independence
      degrades gracefully under loss, whereas V-PCC's inter-frame
      dependencies mean a single lost packet can corrupt an entire
      GOF -- a **loss-amplification** effect.

A realistic deployment: an **edge server** renders the decoded
      volumetric scene and streams the result to a head-mounted
      display over a **private 5G** network. Transport is typically
      **reliable QUIC (HTTP/3-style)**, not a research-modified
      partially-reliable variant. A key consequence: under reliable
      transport, **packet loss is converted into latency** (via
      retransmission), so on such a stack *end-to-end latency* tends
      to dominate perceived-quality degradation rather than raw
      visual corruption. The final display hop from edge renderer to
      HMD (e.g., an ALVR-style cast) behaves as a transparent "
      wireless display" transport and is **not** the volumetric
      streaming path of scientific interest; the interesting network
      effects are observable at the edge application.

Perceived quality (a **Mean Opinion Score**, MOS) can be modeled as
      intrinsic visual quality degraded by network conditions --
      schematically, a visual-quality term (e.g., a point-cloud
      metric such as **PCQM**) attenuated by increasing latency and
      increasing packet-loss rate, with the sensitivity depending on
      the codec's structural resilience. Subjective QoE studies in
      this space follow **ITU-T P.910 (ACR)**. Network impairments
      are commonly emulated with **Linux `tc`** for reproducibility.
      *(These relationships are shape/qualitative grounding only.
      Any coefficients, thresholds, or MOS values you report must
      come from your own experiment, not from this section.)*

### 4. Your mandate -- formulate and commit to a specific
      management contribution

Using the grounding above (but not limited to it), **formulate your
      own specific, falsifiable research question** and a
      corresponding **network/service-management contribution** in
      the 5G/6G + edge domain. You must decide and justify the
      specifics yourself. A strong candidate direction (you may
      adopt, sharpen, or replace it -- but justify your choice):

> **QoE-driven, intent-based / zero-touch closed-loop management of
      volumetric-XR flows over private 5G with edge rendering.** A
      lightweight analytical QoE estimator consumes transport-layer
      telemetry (latency, loss, throughput) as observables and
      drives a control loop that actuates management decisions -- e.
      g., codec-configuration selection, edge-placement / compute
      allocation, or slice-priority adjustment -- to keep MOS above
      an operator-declared *intent* under time-varying network
      conditions. Evaluate the closed-loop policy in simulation
      against static and threshold-based baselines, reporting a
      management-relevant metric such as **intent/SLA-violation rate
      **, MOS stability, or resource efficiency.

Whatever you choose, the paper must make a **generalizable,
      testable claim** -- not merely describe a system. State the
      claim explicitly and show the experiment that could falsify it
      .

### 5. Required paper structure (adapt as needed, but cover all of
      this within 5 pages)

1. **Title, author block** (author: "Agentic AI Researcher";
      affiliation: the CNSM 2026 Agentic AI Researcher Track;
      Research Director named in acknowledgements only), and an **
      abstract** stating the problem, the claim, the method, and the
      headline quantitative result.
2. **Introduction** -- motivation (volumetric XR over private 5G/
      edge is bandwidth-heavy and latency-sensitive; managing its
      QoE autonomously is an open problem), the gap, your specific
      contribution, and an explicit statement of your central claim.
3. **Background & Related Work** -- concise, positioned, and *
      verifiable-references-only*. Distinguish reliable vs.
      partially-reliable QUIC; place your management angle relative
      to existing intent-based / zero-touch / QoE-management work.
4. **System / Method** -- the management architecture and the
      control loop: observables (telemetry), the QoE estimator, the
      actuation space, and how an operator *intent* is expressed and
      enforced. Include one clear architecture figure.
5. **Experimental Setup** -- the simulation you actually ran:
      topology, impairment/traffic model, QoE estimator parameters,
      baseline policies, metrics, and reproducibility details. State
      assumptions honestly.
6. **Results & Discussion** -- figures/tables from your real runs;
      test the central claim; discuss where it holds and breaks;
      state limitations plainly.
7. **Conclusion & Future Work.**
8. **References** -- real and verified only.
9. **Appendix: Disclosure Statement** -- per §1.

### 6. Working process

Think step by step and iterate: (1) commit to the research question
      and central claim; (2) design the simulation and the
      baselines; (3) **write and run the code**, collecting real
      outputs; (4) generate the figures/tables from those outputs;
      (5) write the full paper around them; (6) compile the LaTeX,
      check the ≤5-page limit and reference integrity, and trim.
      Prioritize **soundness and clarity over breadth** -- a small,
      honest, well-executed result beats an ambitious unverifiable
      one for this venue's LLM-as-judge and its adversarial panel.

**Now begin. Produce the complete `IEEEtran` LaTeX source as your
      primary deliverable.**
```

(c) Frameworks and tools. Simulation: Python 3.11 with NumPy (single ∼240-line script, fixed seeds 1000–1029, 2000–2029, 3000–3029), written and executed by the LLM in a sandboxed Linux container. Figures: pgfplots/TikZ compiled from data files emitted by the executed simulation and embedded in this source. Typesetting: IEEEtran with pdfTeX (TeX Live 2023). Reference verification: live web search during the session; every entry in the bibliography was checked against publisher, DOI, RFC-editor, ETSI, or indexing records before inclusion.

(d) Constraints applied. IEEE 2-column conference format, ≤5 pages including references and this appendix; no fabricated testbeds, deployments, subjective studies, or measurements; all reported numbers from actually executed code; verified references only; no reuse of the Research Directors' unpublished results; no human editing of technical content; mandatory verbatim prompt disclosure.

(e) What was and was not automated. Automated by the LLM: problem formulation and the central claim; simulation design, implementation, execution, and statistical analysis; baseline design and tuning; all figures and tables; all prose; LaTeX assembly, compilation, and page-limit iteration; reference search and verification. Not automated: selection of the topic domain and the background context in the prompt above (human Research Directors); the compute environment and this track's rules (organizers). No physical 5G network, HMD, or human rater was involved; the evaluation is exclusively the simulation described in Sec. IV, and its QoE-model parameters are assumptions, not measurements.